Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

MT102T - TWO DIMENSIONAL & THREE DIMENSIONAL GEOMETRY

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define Latus rectum.
2. Write the angle between two plane.
3. Write the equation of the straight line passing through (x_1,y_1).
4. Write the equation of tangent plane.
5. Define cylinder.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find the focus, vertex and the directrix of the parabola $y^2 - 2x - 6y + 5 = 0.$
7. Find the length of perpendicular from origin to the plane $x + 2y + 3z + 4 = 0.$
8. Find the equations to the medians of the triangle whose vertices are (1,2) (2,3) and (-3,-5).
9. Find the equation of the sphere having the circle $x^2 + y^2 + z^2 = 16, x - 2y + 2z = 5$ as a great circle.
10. Find the equation of right circular cone with its vertex at the origin axis along the z=axis and semi vertical angle α.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Show that the locus of the intersection of tangents to $y^2 = 4ax$ which intercept a constant length d on the directrix os $\left(y^2 = 4ax\right)\left(x+a\right)^2 = d^2x^2$.

(or)

b) If the focus, centre and eccentricity of an ellipse are respectively (2,3), (3,4) and ½ find its equation.

12. a) Find the equation of the plane through the point (2,5,-3) and perpendicular to the planes $x+2y+2z=1$ and $x-2y+3z=4$.

(or)

b) Find the equation of the plane passing through the line of intersection $2x+3y-4z=1, 3x-y+z+2=0$ and the point (0,1,1).

13. a) Obtain the co-ordinates of the orthocenter of the triangle whose vertices are the points (1,-2), (3,1) and (-2,3).

(or)

b) Prove that the diagonals of the parallelogram formed by the lines $ax+by+c=0, ax+by+c=0, ax+by+c=0$.

14. a) Find the equation of the tangent planes to the sphere $x^2+y^2+z^2-2x+4y-6z+13=0$.

(or)

b) Find the equation of the sphere for which the circle $x^2+y^2+z^2+7y-2z+2=0, 2x+3y+4z=8$ as great circle.

15. a) Find the equation of the cone generated by the straight line drawn from the origin to cut the circle through the points A(1,0,0), B(0,2,0), C(2,1,1).

(or)

b) Find the equation of right circular cylinder of radius 3 and having for its axis the line $\frac{x-1}{2} = \frac{y-3}{2} = \frac{5-z}{1}$.

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